



EDU TERIA

72nd BPSC

Prelims Online Batch

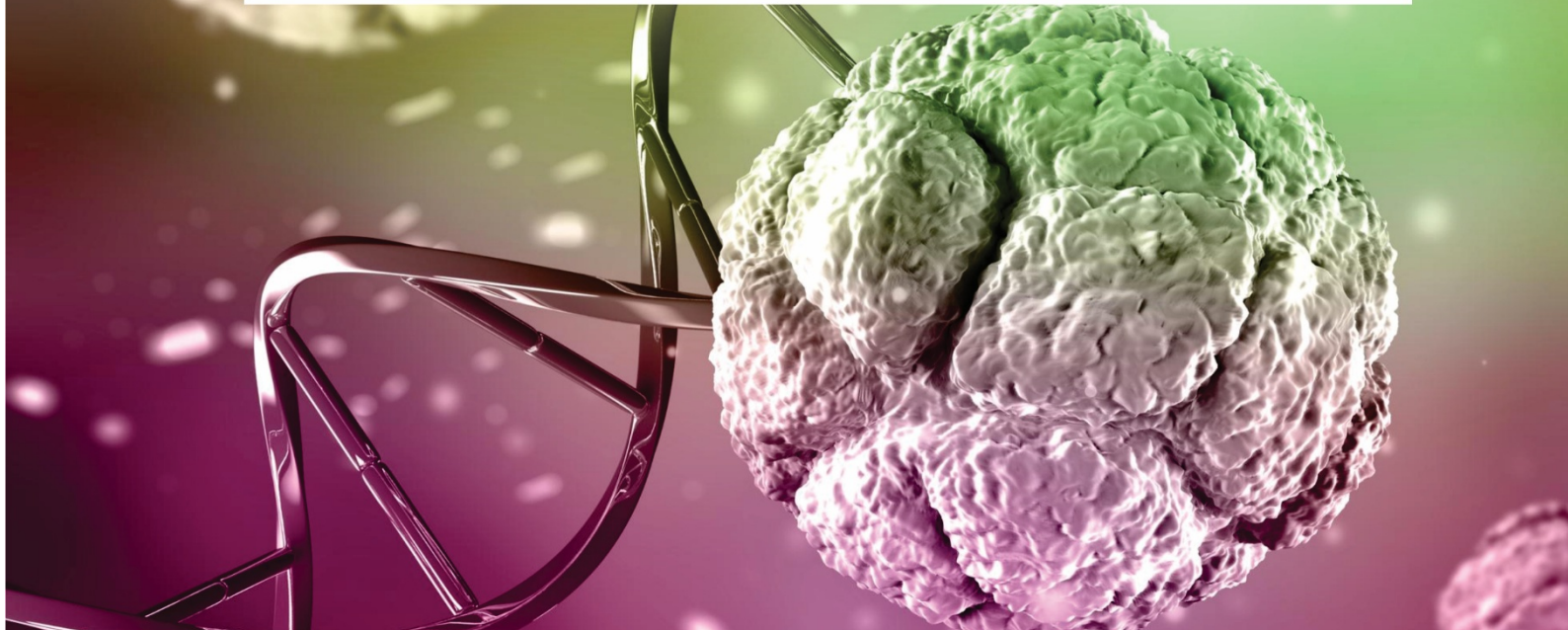
BIOLOGY

Topic wise
Notes

LECTURE

23

Nervous System -02



Nervous System - 02

PERIPHERAL NERVOUS SYSTEM

- All the nerves arising from brain and spinal cord are included in peripheral nervous system. Nerves arising from brain are called cranial nerves, and nerves coming out of spinal cord are called spinal nerves.
- 12-pairs of cranial nerves are found in reptiles, birds and mammals but amphibians and fishes have only 10-pairs of cranial nerves.

SPINAL NERVES

- In Human only 31 pairs of spinal - nerves are found.
- Each spinal nerve is mixed type and arises from the roots of the horns of gray matter of the spinal cord.
- In dorsal root only afferent or sensory fibres and in ventral root efferent or motor fibres are found.
- Both the roots after moving for distance in the spinal cord of vertebrates combine with each other and come out from the Inter vertebral foramen in the form of spinal nerves.

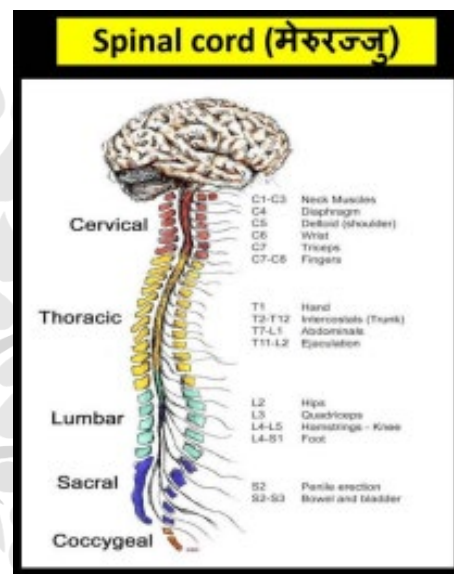
AUTONOMIC OR VISCERAL NERVOUS SYSTEM

- The autonomic nervous system. Visceral nervous system is a part of peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to central nervous system.
- It controls activities inside the body that are normally involuntary, such as heartbeat, peristalsis, sweating etc.

Spinal Cord:

- Extends from medulla to lower vertebral column

- Controls reflex actions and transfers messages between brain and body
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Neuron (Nerve Cell):

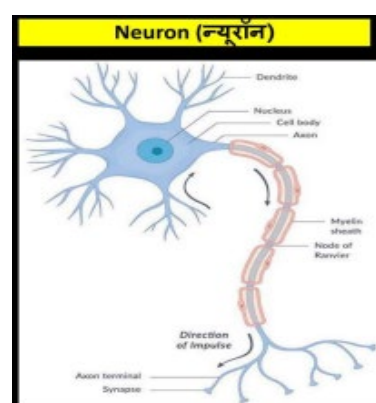
Basic unit of the nervous system.

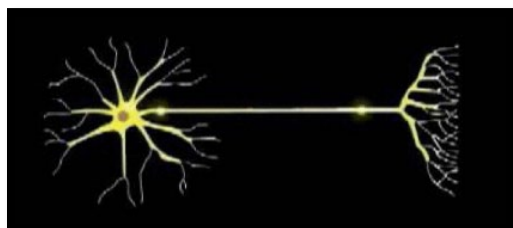
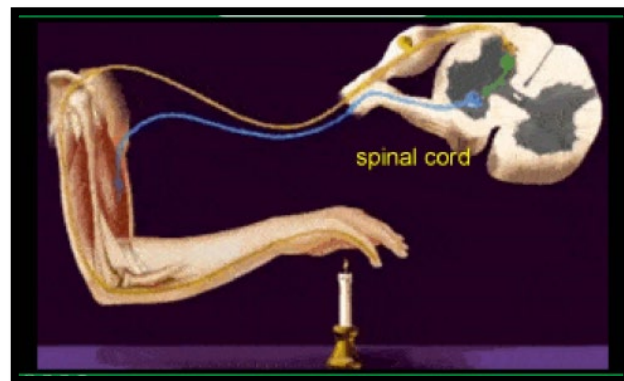
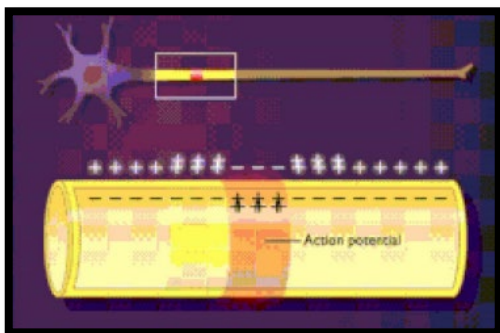
Structure:

- Cell body (cyton)
- Dendrites: Receive impulses
- Axon: Transmits impulses away

Types of Neurons:

Type	Function
Sensory Neuron	Carries message from receptor to CNS
Motor Neuron	Carries message from CNS to effector
Association Neuron	Links sensory and motor neurons





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❑ Reflex Action and Reflex Arc:

- A reflex action is an automatic, quick response that does not involve thinking.
- Examples: blinking, knee jerk, withdrawing hand from flame. Reflex arc sequence:
- Receptor → Sensory neuron → CNS → Motor neuron → Effector (muscle/gland)

❑ Synapse:

- A junction between two neurons
- Transmits message using neurotransmitters (e.g., acetylcholine, dopamine)

❑ Myelin Sheath:

- Fatty covering around axons
- Increases speed of nerve impulse conduction
- Formed by Schwann cells

Disorders of Nervous System

Disorder	Description
Stroke	Brain blood supply blocked; paralysis may occur
Epilepsy	Sudden, recurrent brain activity causing seizures
Parkinson's disease	Lack of dopamine; tremors and stiffness
Alzheimer's disease	Memory loss due to degeneration of brain cells

